

Supplementary Material:

RefCompose: Multi-Reference Image Generation via LoRA-Conditioned Diffusion

Anonymous ECCV 2026 Submission

Paper ID #*****

A Overview

This supplementary document accompanies the main ECCV workshop paper on **RefCompose**. The main paper focuses on Dense Layout quantitative evaluation on InstanceAssemble [8]; here we provide:

1. **Layout-centric benchmarks** (Section B) comparing the prompt-induced spatial layout stage against dedicated layout generators on COCO-GR [3] numerical and spatial reasoning splits.
2. **Extended ablation study** (Section C) with quantitative metrics complementing the qualitative analysis in the main paper.
3. **Training data details** (Section D) referenced from the main experiments section.

Throughout, multiple visual references are composed under a single scene prompt without annotated bounding boxes or per-reference token scaling.

B Layout-Centric Evaluation

We situate the spatial layout stage of RefCompose in the broader layout-generation literature by comparing against LayoutGPT [3] and Lay-Your-Scene [7] on COCO-GR [3] numerical and spatial reasoning splits. RefCompose first induces a coarse composition with a frozen text-to-image model (Flux [2]) and then extracts object boxes with Grounding DINO [6].

Metrics. **L-FID** evaluates layout realism by rendering predicted layouts as colour-coded object maps and computing an FID-style distance to ground-truth layouts [4]; lower is better. For *numerical reasoning*, precision, recall, and accuracy measure whether the predicted object set matches the prompt-specified objects. **Numerical GLIP-Accuracy** checks whether the requested object counts appear in the generated image [5]. For *spatial reasoning*, accuracy measures whether predicted boxes satisfy relations such as *left*, *right*, *above*, or *below*; **spatial GLIP-Accuracy** checks whether those relations remain visible after image generation [5].

Table S1: Layout quality on COCO-GR [3] (lower L-FID is better).

Method	L-FID↓
LayoutGPT [3]	3.51
Lay-Your-Scene [7]	3.07
RefCompose (spatial layout)	2.98

Table S2: Numerical and spatial reasoning on COCO-GR [3]. GLIP-Accuracy measures image-level count/spatial compliance [5].

Method	Numerical Reasoning				Spatial Reasoning	
	Prec.↑	Recall↑	Acc.↑	GLIP↑	Acc.↑	GLIP↑
LayoutGPT [3]	76.29	86.64	76.72	54.25	87.07	56.89
Lay-Your-Scene [7]	77.62	99.23	95.14	56.20	92.58	58.94
RefCompose (spatial layout)	78.21	99.25	96.41	56.29	93.13	59.39

Results. Table S1 reports layout quality; Table S2 reports numerical and spatial reasoning. RefCompose attains the best L-FID and spatial accuracy among compared methods. We attribute this to inverse layout construction: a large pre-trained text-to-image model [2] first produces an aesthetically composed scene, after which boxes are recovered from visible object support. Unlike LLM planners that reason symbolically [3], or layout diffusion transformers trained on narrower layout corpora [7], the T2I prior encodes broad composition statistics from large-scale pretraining; sharper extracted boxes then improve downstream canvas placement and image-level spatial compliance.

C Extended Ablation Study

The main paper presents a qualitative ablation of the reference canvas and depth prior (Figure 7 in the main document). Table S3 reports the corresponding quantitative results on the Dense Layout test split [8], using the same evaluation protocol as Table 2 in the main paper.

Discussion. Removing the *reference canvas* causes the largest drop on Color, Texture, DINO_{refs}, and CLIP-I—appearance-centric signals—while Spatial remains relatively high because the depth prior still constrains placement. The model then infers appearance from text alone, landing near the CreatiLayout/InstanceAssemble range on photometric metrics. CLIP-T increases slightly, mirroring the text-alignment trade-off discussed in the main paper.

Removing the *depth prior* causes the largest drop on Spatial and Shape, reflecting degraded placement and occlusion reasoning, while Color, Texture, and DINO_{refs} remain closer to the full model since the canvas still supplies appearance. This matches the qualitative observation in the main paper that subjects such as the bat and bed remain recognizable, but their arrangement becomes

Table S3: Ablation study evaluating the contributions of the reference canvas and depth prior. The full model consistently achieves the best performance across identity preservation, spatial arrangement, and semantic alignment metrics.

Configuration	Spatial $\uparrow$	Color $\uparrow$	Texture $\uparrow$	Shape $\uparrow$	PICK $\uparrow$	DINO $\uparrow$	DINO _{ref} $\uparrow$	CLIP-I $\uparrow$	CLIP-T $\uparrow$
Full model (canvas + depth, dual LoRA)	0.908	0.826	0.854	0.849	0.229	0.959	0.613	0.766	0.327
w/o reference canvas (text-only appearance)	0.861	0.542	0.571	0.558	0.224	0.712	0.471	0.702	0.334
w/o depth prior (canvas only, no structure)	0.683	0.781	0.809	0.612	0.221	0.798	0.587	0.741	0.310

contextually implausible. Together, these results confirm that canvas and depth provide complementary, disentangled signals for appearance and structure.

D Training Data Details

As noted in the main paper, RefCompose is trained on a mixture of proprietary cinematic imagery and synthetic composites.

Real data. We curate ultra high-resolution movie frames at 1280×720 from a proprietary catalogue. Object references are canonicalized to neutral pose and lighting to reduce spurious correlations between scene context and subject appearance. The five-stage pipeline described in the main paper (Qwen3-VL [1] enumeration, Grounding DINO [6] localisation, Flux [2] canonicalization, DINO filtering, and re-annotation) yields approximately 50,000 samples from 1,000 distinct films.

Synthetic data. We complement real frames with synthetic composites generated using Flux 2.0 [2]. Synthetic scenes emphasise diverse object positions, scales, and orientations that are underrepresented in canonicalized movie crops, while real data supplies richer textures and identity cues. Each synthetic sample follows the same annotation protocol as real data: detected crops, neutral-pose canonicalization, DINO similarity filtering (> 0.90), and dual-level Qwen3-VL [1] captions. This mixture encourages the model to generalise spatial composition beyond studio lighting and fixed poses without sacrificing photometric fidelity on in-domain cinematic content.

Training schedule. LoRA training proceeds in three stages over 50,000 iterations at 1280×720 , as summarised in the main paper: (1) depth-only warm-up on a black canvas (10,000 iterations); (2) appearance grounding with crops placed at Grounding DINO boxes on ground-truth images (20,000 iterations); (3) robustness fine-tuning with photometric jitter and mild spatial perturbations (remaining iterations). Rank $r=32$ low-rank adapters are injected into double-stream attention blocks following EasyControl [9].

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